

Development, simulation & implementation of precoding techniques for satellite links

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Master's dissertation submitted in order to obtain the degree of Master of Science in Computer Science Engineering

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Ghent, May 2026

Wannes Baes

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Faculty of Engineering and Architecture

Department of Telecommunications and Information Processing

Abstract — Linear precoding has become a cornerstone of MIMO communications, as it enables spatial multiplexing gains that translate directly into higher spectral efficiency. These benefits are particularly valuable in multi-beam high-throughput satellite systems, where aggressive frequency reuse across the narrow spot beams turns the forward link into an interference-limited MU-MIMO channel. In this environment, precoding is crucial for mitigating inter-beam interference. In mobile satellite scenarios, however, the combination of long propagation delays and time-varying channels induced by user mobility introduce complications. More specific, the CSI available during the precoder computation is generally outdated with respect to the channel experienced at the time of transmission. This work investigates the performance degradation caused by this CSI mismatch and proposes a low-complexity compensation strategy.

Part I reviews the theoretical foundations underlying this work. After introducing the SU- and MU-MIMO system models, together with the most relevant information-theoretic concepts, a generalised WMMSE framework for joint optimisation of the precoder and combiner in a SU scenario is presented, and the capacity-achieving precoder is derived. For the MU-MIMO broadcast channel, we break down common linear precoding strategies and compare their performance. Although these results are well established, they are revisited here in a unified notation and validated through Monte-Carlo simulations, so that they can serve as a consistent baseline for the satellite-specific analysis that follows.

Part II applies this framework to the forward link of a multi-beam mobile satellite system. The impact of the CSI feedback delay on the system performance is quantified, revealing a substantial performance loss. To counter this degradation, a low-complexity prediction strategy based is proposed. Numerical results show that this scheme recovers a significant fraction of the performance loss under the considered channel model.

Keywords — MIMO, Precoding Techniques, Mobile Satellite Systems, Outdated CSI

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Index Terms—MIMO, Precoding Techniques, Mobile Satellite Systems, Outdated CSI

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ACKNOWLEDGMENT

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Symbols and Notations

Abbreviations

acf	autocorrelation function
AR	autoregressive
AWGN	additive white Gaussian noise
BS	base station
BD	block diagonalization
BER	bit error rate
CS	circularly symmetric
CSI	channel state information
DPC	dirty paper coding
EVD	eigenvalue decomposition
GEO	geostationary earth orbit
GW	gateway
HTS	high-throughput satellite
IBR	information bit rate
i.i.d.	independent and identically distributed
KKT	Karush-Kuhn-Tucker
LEO	low earth orbit
LoS	line-of-sight
LSV	left singular vector
MI	mutual information
MIMO	multiple-input multiple-output
MMSE	minimum mean squared error

MSE	mean squared error
MSS	mobile satellite system
MU	multi-user
NLoS	non-line-of-sight
pdf	probability density function
PSD	power spectral density
QoS	Quality-of-Service
RTT	round-trip time
RX	receiver
SINR	signal-to-interference-plus-noise ratio
SISO	single-input single-output
SNR	signal-to-noise ratio
SR	sum rate
SU	single-user
SVD	singular value decomposition
THP	Tomlinson–Harashima precoding
TX	transmitter
UT	user terminal
WMSE	weighted mean squared error
WMMSE	weighted minimum mean squared error
WSR	weighted sum rate
ZF	zero-forcing

System Parameters

K	Number of UTs
N_T	Number of transmit antennas at the BS
N_R	Number of receive antennas at each UT
N_S	Number of data streams per UT
P_t	Total transmit power
N_0	Noise power on each receive antenna

Symbols

We limit ourselves to the notation for a MU-MIMO downlink system, as this is the most general scenario discussed in this dissertation. The notation for a SU-MIMO system can be obtained by setting $K = 1$ and omitting the user index k .

System Components

$h_{k,(m,n)} \in \mathbb{C}$ — channel coefficient from the n -th transmit antenna of the BS to the m -th receive antenna of user k

$\mathbf{H}_k \in \mathbb{C}^{N_R \times N_T}$ — channel matrix for user k

$\mathbf{H} = [\mathbf{H}_1^T \dots \mathbf{H}_K^T]^T \in \mathbb{C}^{KN_R \times N_T}$ — compound channel matrix

$n_{k,m} \in \mathbb{C}$ — additive noise sample on the m -th receive antenna of user k

$\mathbf{n}_k = [n_{k,1} \dots n_{k,N_R}]^T \in \mathbb{C}^{N_R}$ — noise vector for user k

$\mathbf{n} = [\mathbf{n}_1^T \dots \mathbf{n}_K^T]^T \in \mathbb{C}^{KN_R}$ — compound noise vector

$\mathbf{f}_{k,i} \in \mathbb{C}^{N_T}$ — precoder vector for the i -th data stream intended for user k

$\mathbf{F}_k = [\mathbf{f}_{k,1} \dots \mathbf{f}_{k,N_S}] \in \mathbb{C}^{N_T \times N_S}$ — precoder matrix for user k

$\mathbf{F} = [\mathbf{F}_1 \dots \mathbf{F}_K] \in \mathbb{C}^{N_T \times KN_S}$ — compound precoder matrix

$\mathbf{g}_{k,i} \in \mathbb{C}^{N_R}$ — combiner vector for the i -th data stream intended for user k

$\mathbf{G}_k = [\mathbf{g}_{k,1}^T \dots \mathbf{g}_{k,N_S}^T]^T \in \mathbb{C}^{N_S \times N_R}$ — combiner matrix for user k

$\mathbf{G} = \text{diag}(\mathbf{G}_1, \dots, \mathbf{G}_K) \in \mathbb{C}^{KN_S \times KN_R}$ — compound combiner matrix

System Signals

$a_{k,i} \in \mathbb{C}$ — data symbol in the i -th data stream for user k

$\mathbf{a}_k = [a_{k,1} \dots a_{k,N_S}]^T \in \mathbb{C}^{N_S}$ — data symbol vector for user k

$\mathbf{a} = [\mathbf{a}_1^T \dots \mathbf{a}_K^T]^T \in \mathbb{C}^{KN_S}$ — compound data symbol vector

$x_n \in \mathbb{C}$ — transmit signal on the n -th transmit antenna at the BS

$\mathbf{x} = [x_1 \dots x_{N_T}]^T \in \mathbb{C}^{N_T}$ — transmit signal vector ($\mathbf{x} = \mathbf{F} \mathbf{a} = \sum_{k=1}^K \sum_{i=1}^{N_S} \mathbf{f}_{k,i} a_{k,i}$)

$y_{k,m} \in \mathbb{C}$ — receive signal on the m -th receive antenna of user k

$\mathbf{y}_k = [y_{k,1} \dots y_{k,N_R}]^T \in \mathbb{C}^{N_R}$ — receive signal vector at user k ($\mathbf{y}_k = \mathbf{H}_k \mathbf{x} + \mathbf{n}_k$)

$\mathbf{y} = [\mathbf{y}_1^T \dots \mathbf{y}_K^T]^T \in \mathbb{C}^{KN_R}$ — compound receive signal vector ($\mathbf{y} = \mathbf{H} \mathbf{x} + \mathbf{n}$)

$z_{k,i} \in \mathbb{C}$ — decision variable in the i -th data stream at user k

$\mathbf{z}_k = [z_{k,1} \dots z_{k,N_S}]^T \in \mathbb{C}^{N_S}$ — decision variable vector at user k ($\mathbf{z}_k = \mathbf{G}_k \mathbf{y}_k$)

$\mathbf{z} = [\mathbf{z}_1^T \dots \mathbf{z}_K^T]^T \in \mathbb{C}^{KN_S}$ — compound decision variable vector ($\mathbf{z} = \mathbf{G} \mathbf{y}$)

Part I

Background Material

Part II

Precoding for Mobile Satellite Systems

Part III

Appendices

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